

Exam: Sheet 2
Level:
Course Title: Intelligent Information Systems



Course Code:
Date:
Time:

Q1 Define each of an **organization and **the value chain**.**

Answer:

An **organization** is a group of people that is structured and managed to meet its mission or set of group goals. “Structured” means that there are defined relationships between members of the organization and their various activities, and that processes are defined that assign roles, responsibilities, and authority to complete the various activities. In many cases, the processes are automated using well-defined information systems. Organizations are considered to be open systems, meaning that they affect and are affected by their surrounding environment.

The **value chain** is a series (chain) of activities that an organization performs to transform inputs into outputs in such a way that the value of the input is increased.

Q2 Clarify **Supply chain management (SCM) concept.**

Answer:

Supply chain management (SCM) encompasses all the activities required to get the right product into the right consumer’s hands in the right quantity at the right time and at the right cost—from the identification of suppliers and the acquisition of raw materials through manufacture and customer delivery. The organizations that compose the supply chain are “linked” together through both physical flows and information flows. Physical flows involve the transformation, movement, and storage of supplies and raw materials. Information flows allow participants in the supply chain to communicate their plans, coordinate their work, and manage the efficient flow of goods and material up and down the supply chain.

Q3 Briefly, explain a virtual team.

Answer:

A virtual team is a group of individuals whose members are distributed geographically, but who collaborate and complete work through the use of information systems. The virtual team may be composed of individuals from a single organization or from multiple organizations. The team can consist of just two people up to hundreds of team members. One benefit of virtual teams is that they enable organizations to enlist the best people in different geographical regions to solve important organizational problems. Another benefit is that they provide the ability to staff a team with people who have a range of experience and knowledge that stems from a variety of professional experiences and cultural backgrounds.

Q4 Explain the importance of video conferences in enabling virtual team members to interact with each other.

Answer:

Often, it is difficult for members of a large virtual organization to meet at a time that is convenient for everyone on the team due to the time zone differences in their various geographic locations. Thus, members of a virtual team may seldom meet face to face. However, quick communication exchanges among members can be key to project success. Thus, virtual team members may need to continually monitor their email, instant messages, and team Web site and be prepared to participate in an audio or video teleconference on short notice.

Q5 clarify importance of Change in the Organization.

Answer:

Your organization's current products, services, and ways of accomplishing work are doomed to obsolescence. Fail to change and your competition will take away your customers and your profits. Positive change is a key ingredient for any successful organization. This section will discuss important topics related to change, including innovation, reengineering, continuous improvement, outsourcing, offshoring, and downsizing.

Q6 Clarify **Innovation in an Organization concept.**

Answer:

****Innovation** is the application of new ideas to the products, processes, and activities of a firm, leading to increased value. Innovation is the catalyst for the growth and success of any organization. It can build and sustain profits, create new challenges for the competition, and provide added value for customers. Innovation and change are absolutely required in today's highly competitive global environment; without both, the organization is at risk of losing its competitiveness and becoming obsolete. The following is a list of just a few of today's most innovative products:**

Q7 Briefly, explain each of **Sustaining innovation** and **Disruptive innovation**.

Answer:

Sustaining innovation results in enhancements to existing products, services, and ways of operating. Such innovations are important because they enable an organization to continually increase profits, lower costs, and gain market share.

A **disruptive innovation** is one that initially provides a lower level of performance than the marketplace has grown to accept. Over time, however, the disruptive innovation is improved to provide new performance characteristics, becoming more attractive to users in a new market.

Q8 Briefly, explain each of Outsourcing and Downsizing.

Answer:

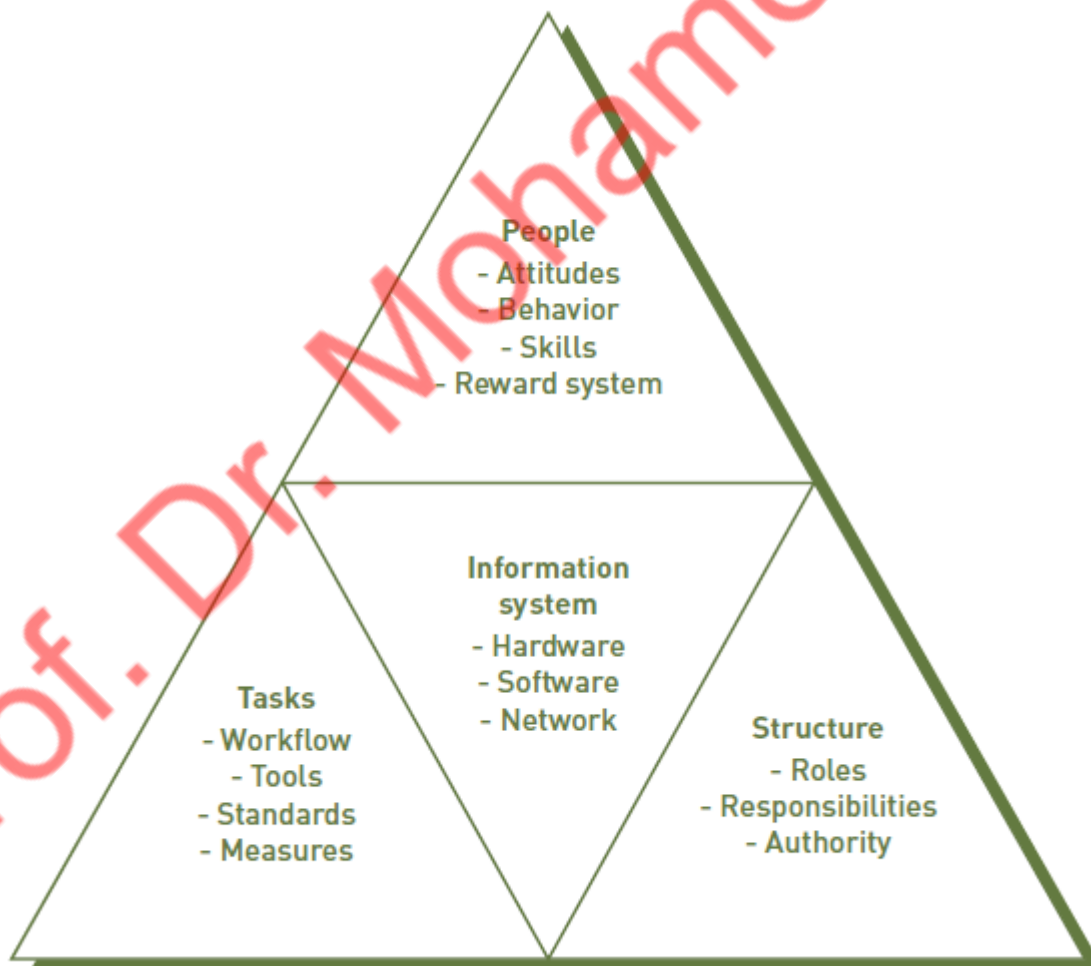
Outsourcing is a long-term business arrangement in which a company contracts for services with an outside organization that has expertise in providing a specific function. Organizations often outsource a process so they can focus more closely on their core business—and target their limited resources to meet strategic goals. Typically, the outsourcing firm has expertise and other resources that enable it to perform the service better, faster, and/or more cheaply. As a result, many companies now outsource jobs such as call center services, payroll activities, information system operations, computer support services, and security services.

Downsizing, a term frequently associated with outsourcing, involves reducing the number of employees to cut costs. The euphemistic term “rightsizing” is sometimes also used. When downsizing, companies usually look to downsize across the entire company, rather than picking a specific business process to downsize. Downsizing clearly reduces total payroll costs, although the quality of products and services and employee morale can suffer. Shortly after Heinz merged with Kraft in March 2015, the food and beverage company announced that it would downsize its 46,000 employees in the United States and Canada by 2,500 people to save \$1.5 billion in annual costs.

Q9 Briefly, explain Leavitt's Diamond and illustrate your answer with a diagram.

Answer:

Leavitt's diamond is a change model that is extremely helpful in successfully implementing change. Leavitt's diamond proposes that every organizational system is made up of four main components—people, tasks, structure, and technology—that all interact; any change in one of these elements will necessitate a change in the other three elements. Thus, to successfully implement a new information system, appropriate changes must be made to the people, structure, and tasks affected by the new system.



Q10 Briefly, explain Diffusion of Innovation Theory.

Answer:

The **diffusion of innovation theory** was developed by E.M. Rogers to explain how a new idea or product gains acceptance and diffuses (or spreads) through a specific population or subset of an organization. A key point of this theory is that adoption of any innovation does not happen all at once for all members of the targeted population; rather, it is a drawn-out process, with some people quicker to adopt the innovation than others.

Adopter Category	Characteristics	Strategy to Use
Innovator	Risk takers; always the first to try new products and ideas	Simply provide them with access to the new system and get out of their way
Early adopter	Opinion leaders whom others listen to and follow; aware of the need for change	Provide them assistance getting started
Early majority	Listen to and follow the opinion leaders	Provide them with evidence of the system's effectiveness and success stories
Late majority	Skeptical of change and new ideas	Provide them data on how many others have tried this and have used it successfully
Laggards	Very conservative and highly skeptical of change	Have their peers demonstrate how this change has helped them and bring pressure to bear from other adopters

With my best wishes

Prof. Dr. Mohamed Okasha